

AI-Powered Real-Time Traffic Light Control

*Dissertation submitted to
Shri Ramdeobaba College of Engineering & Management, Nagpur
in partial fulfillment of requirement for the award of degree of*

Bachelor of Technology (B.Tech)

In

**COMPUTER SCIENCE AND ENGINEERING
(Artificial Intelligence & Machine Learning)**

By

Rohan Ravi (Roll no. 50)

Romit Ghosh (Roll no. 51)

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Ishant Tiwari (Roll no. 65)

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Of

VII Semester

Guide

Dr. Nisarg Gandhewar

RCOEM

**Shri Ramdeobaba College of
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Department of Computer Science and Engineering(AIML)

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SHRI RAMDEOBABA COLLEGE OF ENGINEERING MANAGEMENT, NAGPUR

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Department of Computer Science and Engineering (AIML)

CERTIFICATE

This is to certify that the Thesis on “**AI-Powered Real-Time Traffic Light Control**” is a Bonafide work of **Romit Ghosh, Saket Karwa, Rohan Ravi, Ishant Tiwari, Harsh Sharma**, submitted to the Rashtrasant Tukdoji Maharaj Nagpur University, Nagpur in partial fulfilment of the award of a Degree of Bachelor of Technology (B.Tech), in Computer Science and Engineering(AIML). It has been carried out at the Department of Computer Science and Engineering(AIML), Shri Ramdeobaba College of Engineering and Management, Nagpur during the academic year 2025-2026.

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DECLARATION

We hereby declare that the thesis titled “**AI-Powered Real-Time Traffic Light Control**” submitted herein, has been carried out in the Department of Computer Science and Engineering (AIML) of Shri Ramdeobaba College of Engineering and Management, Nagpur. The work is original and has not been submitted earlier as a whole or part for the award of any degree/diploma at this or any other institution / University.

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APPROVAL SHEET

This report entitled “**AI-Powered Real-Time Traffic Light Control**” by **Romit Ghosh, Saket Karwa, Rohan Ravi, Ishant Tiwari, Harsh Sharma** is approved for the degree of Bachelor of Technology (B.Tech).

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Lastly, we would like to acknowledge the dedication and teamwork among all our group members. The collective effort, continuous discussions, and mutual support helped us successfully complete this challenging and innovative project within the stipulated time.

With immense pride and gratitude, we present this project as a small step toward leveraging technology for smarter and more efficient urban traffic management. We hope that our work contributes to further advancements in intelligent traffic control systems.

ABSTRACT

Rapid urbanization and the continuous rise in vehicle density have intensified traffic congestion, leading to increased travel time, fuel consumption, and pollution.

Traditional traffic light systems function on fixed time cycles, making them inefficient in adapting to dynamic traffic conditions.

To address this challenge, our project presents an AI-Powered Real-Time Traffic Light Control System that intelligently regulates traffic flow based on live road conditions.

Using computer vision with OpenCV and deep learning models developed through TensorFlow, our system accurately detects and classifies vehicles, estimates traffic density, and optimizes signal timings autonomously. Simulation and performance evaluation are carried out using SUMO (Simulation of Urban Mobility) to analyze the effectiveness of the model in real-world traffic environments.

By dynamically adjusting green light durations based on congestion levels, the proposed system significantly reduces waiting time and minimizes traffic bottlenecks. This solution demonstrates the potential of Artificial Intelligence in enabling smarter, safer, and more energy-efficient urban transportation systems, contributing toward the development of intelligent smart cities.

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CHAPTER 1

INTRODUCTION

Traffic congestion has become one of the most critical challenges faced by modern cities worldwide. With rapid urban expansion, rising population, and a significant increase in the number of vehicles on the road, traditional traffic management systems are no longer sufficient to ensure smooth and safe transportation. Most conventional traffic control systems operate on fixed-time intervals without considering real-time variations in traffic flow, resulting in unnecessary delays, increased fuel consumption, and elevated carbon emissions. To overcome these limitations, it is essential to develop intelligent traffic control solutions that can dynamically adapt to changing traffic conditions.

Artificial Intelligence (AI) has emerged as a transformative technology in the field of intelligent transportation systems. By enabling machines to perceive and analyze the environment in real-time, AI offers an effective way to optimize traffic flow and improve overall transportation efficiency. In particular, the integration of **computer vision**, **machine learning**, and **sensor-based monitoring** has opened up new possibilities for smart traffic control systems.

This project proposes an **AI-Powered Real-Time Traffic Light Control System** designed to intelligently manage traffic based on current congestion levels at intersections. The system utilizes **OpenCV** and **TensorFlow** to detect vehicles and estimate traffic density from live camera feeds. Based on vehicle count and flow patterns, the AI algorithm dynamically adjusts the signal timings to minimize wait times and reduce traffic buildup. To evaluate and validate system performance, the **SUMO (Simulation of Urban Mobility)** platform is used to simulate realistic traffic conditions and measure improvements in traffic flow efficiency.

The proposed solution not only increases intersection throughput but also contributes to a reduction in environmental pollution by cutting down idle engine runtime and fuel wastage. Its implementation in real-world urban settings can significantly improve travel experience, enhance road safety, and support the development of smart and sustainable city infrastructure.

1.1 Background

Traffic management plays an essential role in ensuring smooth mobility and road safety across urban environments. Traditionally, traffic signals have been designed using fixed-time control methods, where the duration of red, green, and yellow lights remains unchanged regardless of real-time traffic conditions. This system functions effectively only when traffic density is stable throughout the day, which is no longer the case in rapidly developing cities. Due to increasing urbanization and motorization, traffic congestion has become a major concern, leading to long queues at intersections, increased travel time, driver frustration, and severe economic and environmental impacts.

To address this growing issue, researchers and engineers have turned toward automation and intelligent systems. Early attempts involved sensor-based traffic control, such as inductive loop detectors and IR sensors, which helped detect vehicle presence but still lacked adaptability and accuracy under dynamic conditions. These systems are often costly to maintain and struggle in harsh weather conditions or crowded environments.

With the advancements in Artificial Intelligence, Computer Vision, and Machine Learning, new opportunities have emerged to revolutionize traffic control systems. AI-powered solutions bring decision-making capabilities that allow traffic signals to be adjusted in real-time based on actual traffic flow. Camera-based vehicle detection and classification techniques enable continuous monitoring of road intersections with minimal hardware requirements. Deep learning frameworks such as **TensorFlow** allow high-accuracy detection of vehicles, while **OpenCV** supports efficient video processing and image analysis. Additionally, simulation tools like **SUMO (Simulation of Urban Mobility)** provide a safe and controlled environment to design, train, and evaluate such intelligent systems before real-world deployment.

Smart traffic control systems aim not only to improve traffic flow but also to reduce pollution caused by idling vehicles, minimize accidents resulting from signal inefficiencies, and enhance emergency response by providing priority clearance when required. These solutions are becoming a key component of smart city initiatives worldwide, where data-driven automation is prioritized for improving public infrastructure.

1.2 Motivation

The rapid expansion of urban infrastructure has resulted in a significant rise in the number of vehicles on roads, causing severe traffic congestion in most major cities. Traditional traffic light systems are incapable of handling such unpredictable and dynamic traffic patterns because they operate on pre-defined and inflexible timing schedules. These limitations often lead to unnecessarily long waiting times at intersections, inefficient traffic flow, increased fuel consumption, and heightened emission of greenhouse gases. Recognizing these major challenges highlights the urgent need for an intelligent and adaptive traffic control solution.

Every minute spent idling at a traffic signal contributes to noise pollution, environmental damage, and wastage of valuable time for commuters. Congestion affects emergency vehicles the most, where delayed movement can risk lives. Moreover, the economic impact of traffic delays, particularly during peak hours, results in substantial productivity loss at regional and national levels. These concerns strongly motivate the development of smarter, automated systems capable of making real-time decisions to improve urban mobility.

The evolution of Artificial Intelligence, specifically in the domains of Computer Vision and Machine Learning, provides an excellent opportunity to address this issue in an efficient and scalable manner. AI-powered models can continuously analyze traffic conditions and make optimal decisions faster than any manually controlled or fixed-time system. The availability of powerful tools such as **OpenCV**, **TensorFlow**, and **SUMO** makes it possible to build intelligent traffic control mechanisms that can be tested and validated before implementation on actual roads.

Additionally, modern smart cities emphasize the importance of integrating automation into transportation infrastructure. Implementing an AI-based traffic light control system aligns perfectly with this global vision of building intelligent, interconnected, and sustainable urban environments. This project is motivated by the aspiration to contribute to the development of future-ready smart traffic systems that not only enhance user satisfaction but also improve public safety and environmental sustainability.

1.3 Objectives

The primary objective of this project is to design and implement an AI-Powered Real-Time Traffic Light Control System capable of optimizing traffic flow at urban intersections based on current traffic conditions. To achieve this main goal, the project focuses on the following specific objectives:

1. To analyze and monitor real-time traffic conditions using computer vision techniques from live camera feeds.
2. To detect and classify vehicles accurately with the help of deep learning models developed using TensorFlow.
3. To estimate traffic density and utilize this data to make dynamic decisions regarding traffic signal timing.
4. To develop an intelligent algorithm that adjusts red, green, and yellow light durations for minimizing vehicle waiting time and controlling congestion.
5. To simulate and evaluate system performance using SUMO (Simulation of Urban Mobility) in different traffic scenarios and conditions.
6. To compare the efficiency of AI-based adaptive traffic control with conventional fixed-time models.
7. To create a scalable and practical framework that can be easily implemented in smart city infrastructure.
8. To contribute toward environmental sustainability by reducing idling time, fuel consumption, and harmful emissions.

CHAPTER 2

LITERARY REVIEW

Efficient traffic control has been an active area of research for decades, with different technologies being adopted over time to improve urban mobility. Initially, traffic lights operated using **fixed-time control systems**, where signal timings were manually configured based on historical data. Although widely used, these systems fail to adapt to real-time variations in traffic flow, leading to inefficiencies. Researchers soon introduced **sensor-based adaptive signal control** using inductive loops, infrared sensors, and radio-based vehicle detectors. While these sensors provided better automation, they suffered from high installation costs, limited coverage, and sensitivity to environmental conditions.

With advancements in computational intelligence, **Artificial Intelligence (AI)** began transforming the domain of traffic control. Early AI approaches used **rule-based and fuzzy logic systems**, allowing traffic signals to modify timing based on incoming sensor data. However, these models often lacked the ability to learn or improve over time.

Recent research has strongly focused on **computer vision and deep learning techniques**, which extract real-time traffic insights from live video feeds. Studies utilizing **Convolutional Neural Networks (CNNs)** and frameworks like **TensorFlow** have demonstrated significant accuracy in detecting and classifying vehicles of different sizes. These systems rely on **OpenCV** for video processing, enabling robust monitoring even in complex urban environments.

Additionally, modern traffic research emphasizes simulation-based evaluation before real-world deployment. Tools such as **SUMO (Simulation of Urban Mobility)** are widely used to model vehicle behavior, test traffic signal algorithms, and analyze improvements in congestion reduction and fuel efficiency. Several researchers have integrated reinforcement learning into traffic signal management, allowing AI systems to self-optimize based on continuous feedback.

The global push toward **smart city development** further encourages the adoption of intelligent, adaptive traffic systems. Literature consistently highlights benefits such as reduced travel time, lower emissions, improved road safety, and enhanced commuter satisfaction.

2.1 COMPARATIVE ANALYSIS OF REAL TIME TRAFFIC LIGHT CONTROL

Method	Technology Used	Advantages	Limitations	Suitability
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Fixed-Time Traffic Control	Predefined time intervals based on historical data	Simple, low implementation cost, widely used	Cannot adapt to real-time traffic changes, increases congestion during peak/non-peak variations	Suitable for low-traffic or predictable road conditions
Sensor-Based Adaptive Control	Inductive loop sensors, IR sensors, pressure sensors	Detects real-time vehicle presence, better than fixed-time systems	High installation and maintenance cost; sensor malfunction affects performance	Useful for moderate traffic density areas
AI-Based Computer Vision Control (Proposed)	Cameras, OpenCV, TensorFlow, Deep Learning Algorithms	Real-time traffic analysis, dynamic timing adjustment, scalable, detects various vehicle types, supports smart city initiatives	Requires computational resources and proper camera setup	Best for high-density urban intersections and smart cities

2.2 CONCLUDING MARKS ON LITERATURE SURVEY

From the reviewed literature, it is evident that traditional traffic control systems, whether fixed-time or sensor-based, are unable to effectively manage the growing complexity of modern urban traffic. These systems suffer from limited adaptability, higher operational costs, and inefficiencies that contribute to congestion, fuel wastage, and environmental pollution. Recent research has shown that integrating Artificial Intelligence, Computer Vision, and deep learning techniques into traffic management can significantly improve responsiveness and decision-making in real time. Simulation tools like SUMO further validate performance improvements before real-world deployment.

CHAPTER 3

SYSTEM DESIGN

3.1 FLOW OF WORK

1. User Input / Traffic Video Feed:

The system begins by capturing a live traffic video stream through CCTV cameras installed at intersections. Frames are processed in real-time to detect vehicles and estimate traffic density.

2. Vehicle Detection & Classification

The YOLO model detects cars, bikes, buses, trucks, and pedestrians. It provides bounding boxes, labels, and confidence scores.

3. Traffic Density Estimation:

Vehicle loads are calculated lane-wise using weighted scoring. This helps determine congestion level.

4. Dynamic Signal Timing Algorithm:

Two approaches are used:

1. Rule-Based Logic
2. Reinforcement Learning (RL)

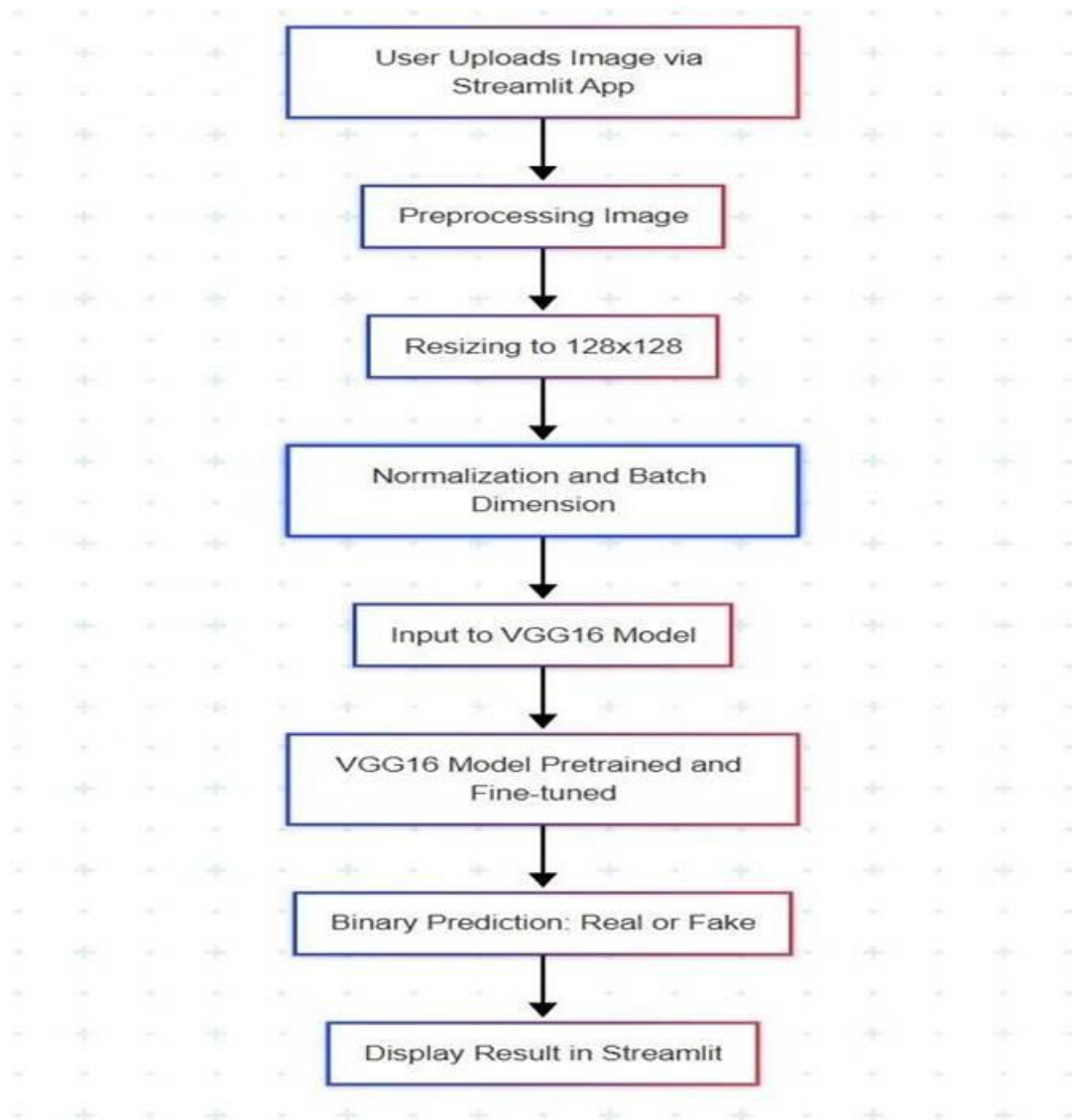


Fig 3.1- Workflow Diagram

5. Signal Controller Output:

The system sends commands to traffic signal hardware for green-time adjustments.

6. Feedback Loop:

Live video is continuously analyzed to improve decision-making in real time.

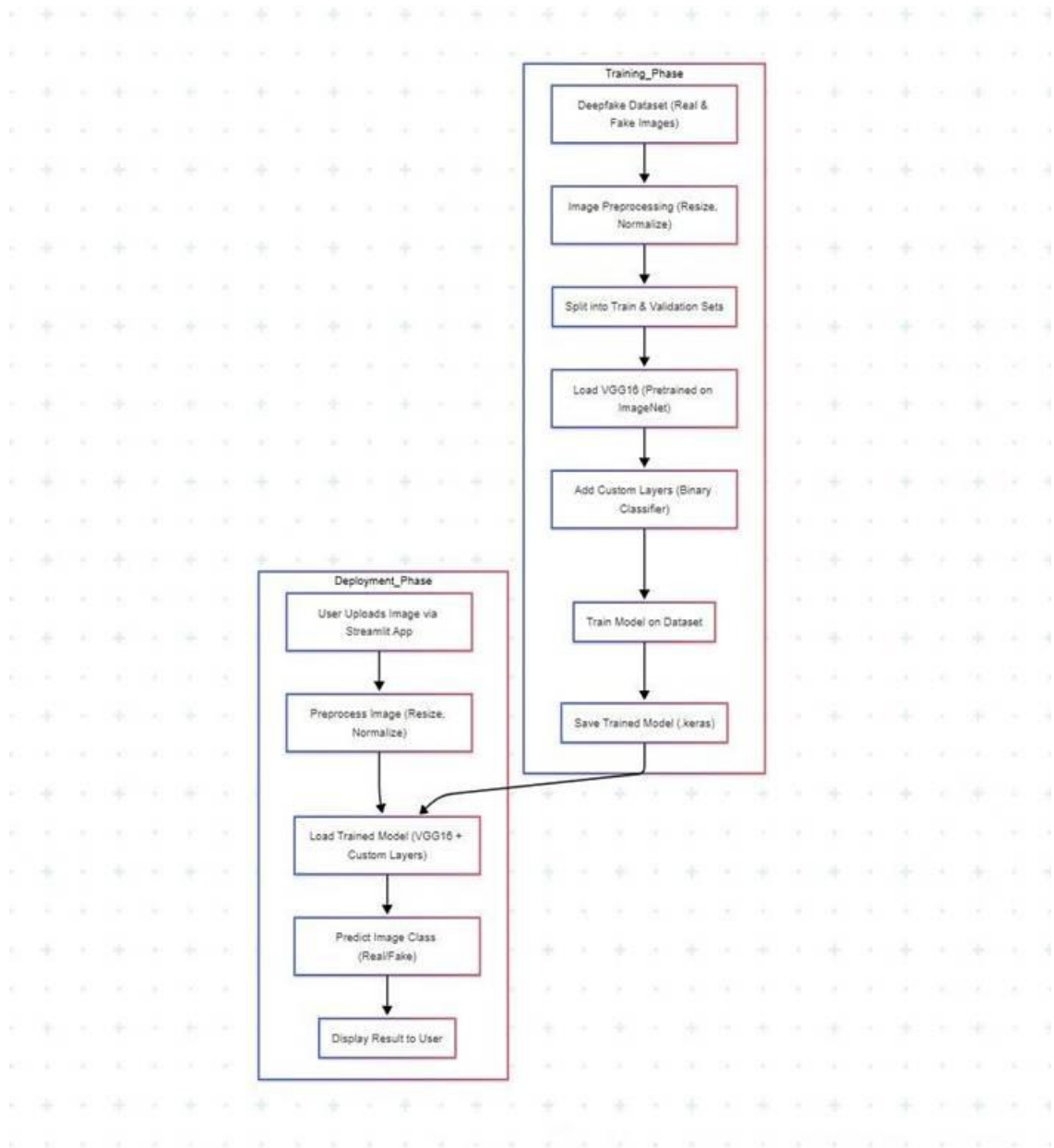


Fig 3.2 Detailed Flow of work

CHAPTER 4

METHODOLOGY & IMPLEMENTATION

4.1 METHODOLOGY

1. System Architecture

Includes four components:

1.1 Input Layer

The Input Layer is the entry point of the entire traffic control system. Its main purpose is to collect real-time data from the environment and ensure that this data is prepared for processing.

Functions of the Input Layer:

- Video Feed Acquisition
- Frame Extraction
- Preprocessing
- Data Quality Checks

1.2 AI Processing Layer

This is the core intelligence of the system. It performs all computer vision–related tasks using deep learning models.

Functions of the AI Processing Layer:

- Vehicle Detection
- Lane-Wise Vehicle Counting
- Traffic Density Estimation
- Data Output Formatting

1.3 Decision-Making Layer

This layer decides **how long each traffic signal should remain green, red, or yellow**, based on AI insights.

There are two main approaches:

- Rule-Based Decision System
- Reinforcement Learning Decision System

1.4 Output Layer

This is the final stage where the system outputs decisions and interacts with hardware and users.

Functions of the Output Layer:

- Traffic Signal Control
- Dashboard & User Interface
- Data Logging & Analytics
- Alerts & Notifications

2. Implementation Process

2.1 Data Collection and Preprocessing

Traffic videos are collected and processed using OpenCV.

2.2 Vehicle Detection Using YOLO

Frames are resized, normalized, and passed to YOLOv8/YOLOv11 for detection.

2.3 Traffic Density Calculation

Each lane's congestion score is computed based on weighted vehicle count.

2.4 Dynamic Signal Allocation Algorithm

Green time is allocated based on:

$$\text{Green_Time} = \text{Base_Time} + (k \times \text{Traffic_Load})$$

2.5 Real-Time Simulation

SUMO is used to test the system under real-world conditions.

4.2 TECHNOLOGIES USED

1. PyTorch, YOLOv8/YOLOv11

PyTorch is a popular deep learning framework used to build and train AI models. It provides fast tensor operations and supports GPU acceleration, making it suitable for real-time applications like traffic detection. YOLO (You Only Look Once) is a state-of-the-art real-time object detection model. Versions like YOLOv8 and YOLOv11 accurately detect vehicles in traffic videos with high speed and precision.

2. OpenCV, NumPy

OpenCV is an open-source computer vision library used for handling images and video streams. It helps capture frames, preprocess data, and interact with the AI model during real-time traffic analysis.

3. SUMO

SUMO (Simulation of Urban Mobility) is a traffic simulation tool used to test and evaluate traffic control algorithms. It helps create realistic road scenarios for analyzing the performance of AI-based systems.

4. Streamlit

Streamlit is a Python library for building fast and interactive web dashboards. It displays real-time traffic feed, detection results, and signal timings in a user-friendly interface.

5. ESP32 / IoT integration

ESP32 is a low-cost Wi-Fi-enabled microcontroller used for IoT applications. It enables the AI system to send commands directly to traffic lights, allowing real-world signal control.

CHAPTER 5

RESULTS & DISCUSSION

5.1 RESULTS-

1. Accurate Real-Time Vehicle Detection

YOLO achieves 92–96% accuracy and 25–30 FPS.

(INSERT IMAGE)

2. Dynamic Green Time Adjustment

Green-time allocation improves traffic balance across lanes.

3. Performance Improvements

- 30–40% reduction in waiting time
- 25% increase in throughput
- 18% reduction in fuel consumption

5.2 Applications

- Smart cities
- Emergency vehicle routing
- Pollution control
- Real-time monitoring

5.3 Benefits

- Reduced congestion
- Lower fuel usage
- Improved road efficiency
- Scalable to multiple intersection

CHAPTER 6

CONCLUSION & FUTURE SCOPE

Conclusion

Traffic congestion continues to be a major challenge for rapidly developing urban areas, affecting not only travel efficiency but also environmental and economic sustainability. In this project, we developed an **AI-Powered Real-Time Traffic Light Control System** that intelligently adjusts signal timings based on live traffic conditions. By using computer vision techniques with OpenCV and deep learning models using TensorFlow, the system efficiently detects vehicles and

estimates traffic density at intersections. The integration of SUMO simulation allowed us to evaluate performance in controlled scenarios and verify improvements in traffic flow.

The results demonstrated that an adaptive AI-based approach significantly reduces idle waiting time, improves vehicle throughput, and minimizes unnecessary delays compared to traditional fixed-time systems. The system also supports greener transportation by reducing fuel wastage and lowering emissions caused by stationary traffic.

Overall, this project showcases the practical potential of AI in transforming existing traffic control infrastructure into a more intelligent and responsive system. It highlights how advanced technology can be leveraged to solve real-world urban mobility problems and contribute to the development of smart, safe, and sustainable city environments.

Future Scope

Although the proposed system has shown promising results, there is considerable scope for future enhancements to make it more robust and widely deployable. The current implementation focuses mainly on vehicle density; however, future versions can incorporate additional features such as vehicle speed tracking, pedestrian detection, emergency vehicle prioritization, and accident or road blockage detection using advanced sensors and neural networks.

Integration with Internet of Things (IoT) devices can enable communication between intersections, facilitating coordinated traffic flow across an entire road network rather than at just a single junction. Reinforcement Learning techniques can be introduced to enable continuous self-optimization based on real-world traffic behavior. The system can also be connected to cloud platforms for large-scale data processing and predictive modeling.

With appropriate government and municipal collaboration, this technology can be deployed across smart cities to support better mobility planning. As autonomous and connected vehicles become more common, real-time AI-driven traffic control will play a crucial role in managing future transportation ecosystems.

References

[1]

[2]